CONTROL SYSTEMS 354 · TEST

# Memo and Solution Framework

Total **80 marks** Walkthrough **9 Oct 2026** Reconciled **13 Oct 2026**

> Built from the lecturer's memo walkthrough and two independent transcriptions of it. Every number below was checked by working the problem through. Marker's notes are the lecturer's own comments on how marks were given.

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### ANSWERS AT A GLANCE

| PART | MARKS | RESULT |
|---|---|---|
| 1(a)(i) | 6 | $y_{ss} = 1/3$; requirement not met |
| 1(a)(ii) | 6 | Poles $-1 \pm j3.74$; need $\sigma \geq 4$; not met |
| 1(b) | 4 | $K = 3$; no feedback to correct the output |
| 1(c) | 10 | PI or PID; integral action makes the loop type 1 |
| 2(a) | 6 | PD raises high-frequency gain (bandwidth) and phase (margin) |
| 2(b) | 20 | $D(s) = 20\,\dfrac{0.51s + 1}{0.0102s + 1}$ |
| 3.1 | 8 | Pole angles sum to $-180°$; $s_0$ is on the locus; $L(s) = G(s)$ |
| 3.2 | 10 | $K = 5$; $e_{ss} = 0.4 > 0.01$; not met |
| 3.3 | 10 | $\alpha = 40$; $D(s) = 5\,\dfrac{s + 0.224}{s + 0.0056}$ |

## Question 1 · Open loop and PID selection [26]

$$G(s) = \frac{5}{s^2 + 2s + 15}, \qquad \text{require } e_{ss} = 0 \text{ (unit step)}, \quad T_s < 1\text{ s } (2\%)$$

**(a)(i)** *Open loop, no control: find $y_{ss}$ for a unit step.* (6)

1. **Write the output, not the error.**

$$Y(s) = G(s)\,R(s), \qquad R(s) = \frac{1}{s}$$

2. **Apply the final value theorem.**

$$y_{ss} = \lim_{s \to 0} s\,G(s)\,\frac{1}{s} = G(0) = \frac{5}{15} = \frac{1}{3}$$

3. **State the implication.**

A unit step should give $y_{ss} = 1$, so $e_{ss} = 1 - \frac{1}{3} = \frac{2}{3} \neq 0$.

**ANSWER**

$y_{ss} = 1/3$. The steady-state requirement is not met.

**MARKER'S NOTES**

- Marks for the correct FVT expression, defining $Y(s)$ and $R(s)$, the value, and the implication.
- Most common error: applying the FVT to $E(s)$ instead of finding $y_{ss}$.

**(a)(ii)** *Plot the poles and the settling time region.* (6)

1. **Poles.**

$$s^2 + 2s + 15 = 0 \;\Rightarrow\; p_{1,2} = -1 \pm j\sqrt{14} \approx -1 \pm j3.74$$

2. **Settling time boundary**

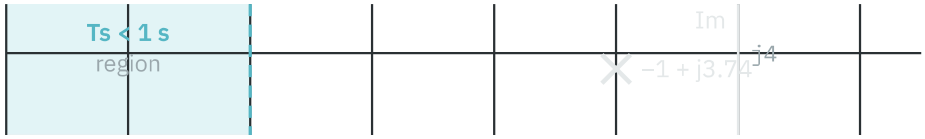
from the formula sheet (2%):

$$T_s = \frac{4}{\sigma} < 1 \;\Rightarrow\; \sigma > 4$$

Poles must lie to the left of $\mathrm{Re}(s) = -4$.

3. **Compare.**

The poles sit at $\sigma = 1$, giving $T_s \approx 4$ s.



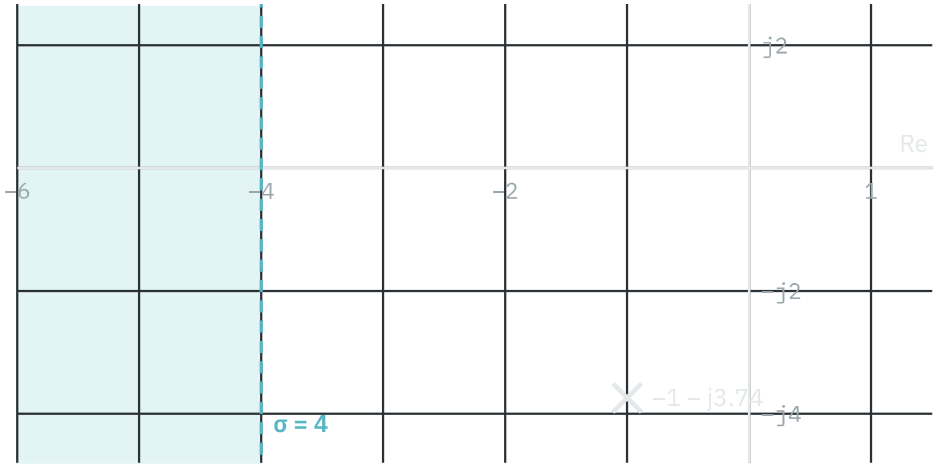



**Figure 1.** Open-loop poles against the 2% settling time region (shaded, $\sigma > 4$).

**ANSWER**

Poles at $-1 \pm j3.74$ lie outside the region $\mathrm{Re}(s) < -4$. The settling time requirement is not met.

**MARKER'S NOTES**

- The sketch alone was not enough: you had to state that the poles are outside the region (or must be moved left) to justify the answer.

(b) *Proportional gain in series, open loop.* [4]

$$y_{ss} = \lim_{s \to 0} s\, K\, G(s)\, \frac{1}{s} = \frac{K}{3} = 1 \;\Rightarrow\; K = 3$$

**ANSWER**

$K = 3$. Disadvantage: there is no feedback, so the controller never sees the actual output and cannot correct for disturbances or model error.

(c) *Unity feedback, choose from P, I, D using system type.* [10]

1. **Write the error in general feedback form.**

$$E(s) = R(s) - Y(s) = \frac{R(s)}{1 + D(s)G(s)}$$

2. **No controller (or P only): type 0.**

$$e_{ss,\text{step}} = \frac{1}{1 + K_p},\; K_p = \lim_{s \to 0} D(s)G(s) \neq 0, \qquad e_{ss,\text{ramp}} \to \infty$$

With $D = 1$: $K_p = 1/3$ and $e_{ss} = 0.75$.

3 **Add integral action: type 1.**

$$D(s) = K_P + \frac{K_I}{s} \Rightarrow e_{ss,\text{step}} = 0, \qquad e_{ss,\text{ramp}} = \frac{1}{K_v} = \frac{3}{K_I}$$

4 **Transient.**

Proportional and/or derivative action is needed to move the dominant poles left of $-4$ for $T_s < 1$ s.

**ANSWER**

PI or PID control. The integrator raises the loop to type 1 for zero step error; P (and D) shape the transient.

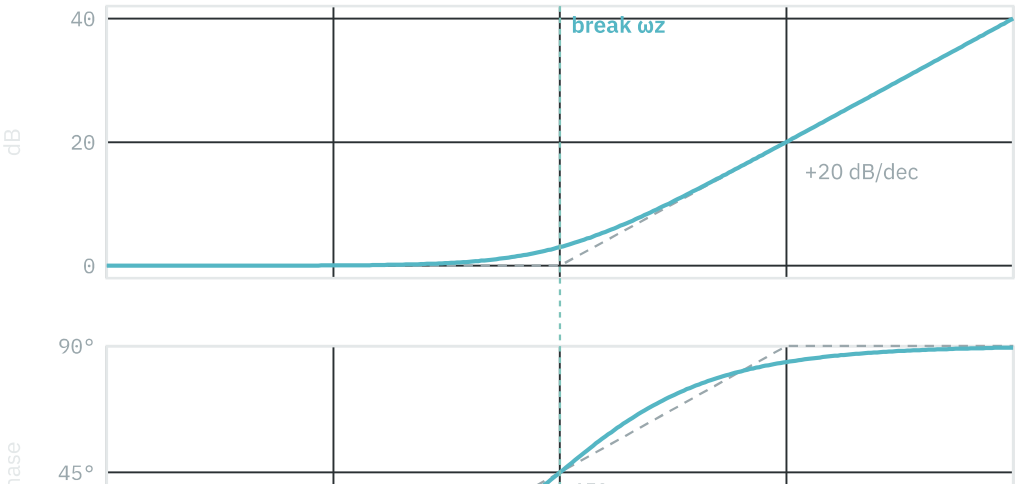
**MARKER'S NOTES**

- 2 marks for $E(s)$ written as $R - Y$ together with the expression for $Y(s)$.
- Applying the controller first and then checking the type was also accepted; the question was open-ended.
- P alone stays type 0: process marks but not the recommendation mark. I alone meets tracking but has poor transient performance: also loses the recommendation mark.

## Question 2 · Frequency response design [26]

Plant given as an open-loop Bode plot: type 1 with $\lim_{s\to 0} sG(s) = 1$. Require $e_{ss} \le 0.05$ (unit ramp) and $\%OS < 5\%$ (unit step).

**(a)** *Benefits of PD control from its Bode plot.* [6]



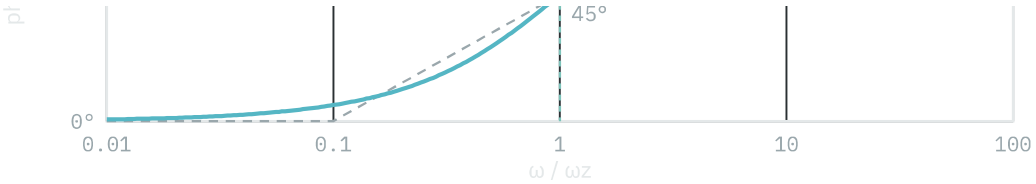



Figure 2. PD controller $D(s) = K_P(1 + s/\omega_z)$, normalised to $K_P = 1$. Dashed lines are the asymptotes.

- **Magnitude.** Flat below the break frequency $\omega_z$, then rising at +20 dB/decade. Higher gain at high frequency pushes crossover out, increasing bandwidth, so the transient response is faster.
- **Phase.** Steps up from 0° towards +90° (45° at $\omega_z$). The added phase lead raises the phase margin, making the system more stable with less overshoot.

#### MARKER'S NOTES

- Marks for showing a break frequency (exact labelling not penalised), the gain increase and its effect on bandwidth, and the phase increase and its effect on phase margin.

(b) *Lead compensator* $D(s) = K\,\dfrac{Ts + 1}{\alpha T s + 1}$ *by frequency response.* [20]

1 **Gain from the steady-state error (plant alone first).**

$$e_{ss} = \lim_{s\to 0} s\,\frac{1/s^2}{1 + KG(s)} = \frac{1}{K\lim_{s\to 0} sG(s)} = \frac{1}{K} \le 0.05 \;\Rightarrow\; l$$

$$20\log_{10} 20 \approx 26 \text{ dB}$$

2 **New crossover.**

Shift the magnitude up by 26 dB, i.e. find where $|G| = -26$ dB on the plot: $\omega_c \approx 14$ rad/s. The phase there is close to $-180°$, so $\phi_{\text{current}} \approx 0°$.

3 **Phase lead needed.**

From the overshoot figure, $\%OS < 5\%$ gives $\zeta \approx 0.69$ and $\phi_{\text{req}} \approx 65°$. Adding about 10° for safety:

$$\phi_{\max} = \phi_{\text{req}} - \phi_{\text{current}} + \phi_{\text{safety}} \approx 65° - 0° + 10° = 75°$$

4 **$\alpha$ from $\phi_{\max}$.**

Reading the formula sheet figure at 75° gives $1/\alpha \approx 50$, so

reading the formula sheet figure at 75° gives $1/\alpha \approx 58$, so $\alpha \approx 0.02$. The exact relation gives a similar value:

$$\alpha = \frac{1 - \sin\phi_{\max}}{1 + \sin\phi_{\max}} = 0.017 \quad (1/\alpha \approx 58)$$

5 **$T$ by placing $\phi_{\max}$ at the new crossover.**

$$\omega_m = \frac{1}{T\sqrt{\alpha}} \Rightarrow T = \frac{1}{14\sqrt{0.02}} \approx 0.51 \text{ s}, \qquad \alpha T \approx 0.0102 \text{ s}$$

6 **Substitute everything.**

$$D(s) = 20\,\frac{0.51\,s + 1}{0.0102\,s + 1}$$

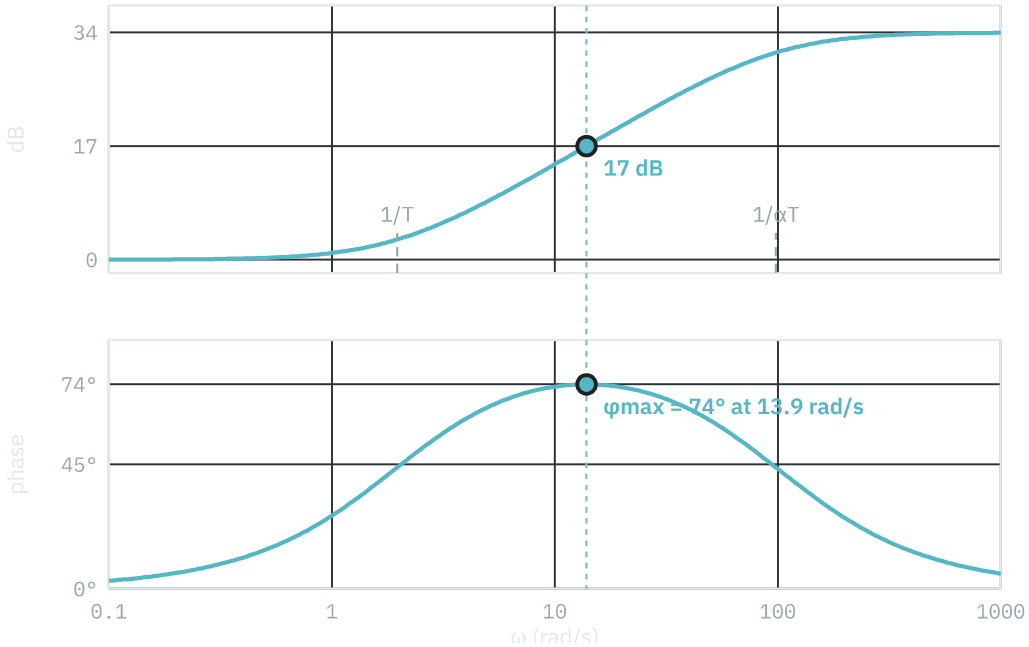



**Figure 3.** The designed lead network $(0.51s + 1)/(0.0102s + 1)$, gain $K$ excluded. Peak phase 74° at 13.9 rad/s, where the network adds 17 dB.

**ANSWER**

$K = 20$, $\phi_{\max} \approx 75°$, $\alpha \approx 0.02$, $T \approx 0.51$ s, giving $D(s) = 20\,(0.51s + 1)/(0.0102s + 1)$.

**MARKER'S NOTES**

- Plot readings had a broad tolerance (crossover read anywhere in a sensible band was accepted if explained).
- A wrong crossover cost that mark and the downstream final-answer marks ($\phi_{\text{current}}$, $\phi_{\max}$, $T$); method marks for each equation were kept.
- Final mark: all values substituted into $D(s)$. Not showing the 26 dB magnitude shift was marked wrong.

**CHECK**

The memo places $\omega_m$ at the gain-shifted crossover. A stricter textbook method places it where $|KG| = -10\log_{10}(1/\alpha) \approx -17$ dB, since the network itself adds 17 dB there. Follow the memo for this test.

## Question 3 · Root locus design [28]

$$G(s) = \frac{1}{s(s+2)}, \qquad s_0 = -1 \pm j2, \qquad K_v = 100 \ (e_{ss} = 0.01, \ \text{unit ramp})$$

**3.1** *Show the locus passes through $s_0$ and identify $L(s)$.* [8]

1. **Root locus form.**

   With proportional control the characteristic equation is

   $$1 + KG(s) = 0 \quad \equiv \quad 1 + KL(s) = 0 \ \Rightarrow \ L(s) = G(s)$$

2. **Angle condition at $s_0 = -1 + j2$.**

   No zeros, so the zero angle sum is 0.

   $$\angle L(s_0) = 0 - \left(\theta_0 + \theta_{-2}\right) = -\left(116.6^\circ + 63.4^\circ\right) = -180^\circ$$

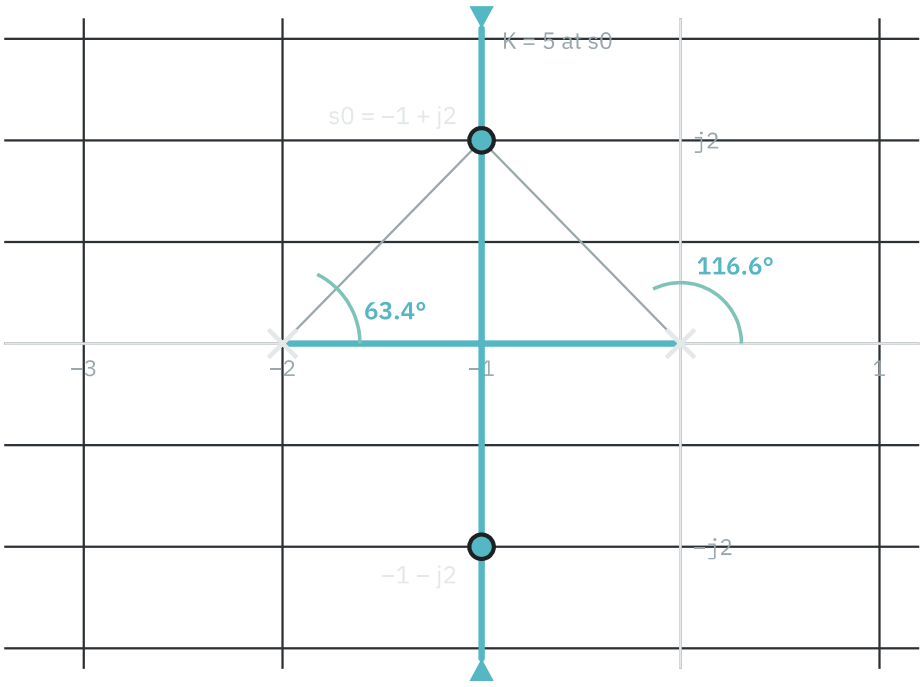



Figure 4. Root locus of $1/(s(s+2))$: the branches meet at $-1$ and run vertically

passing through $s_0 = -1 \pm j2$.

**ANSWER**

The angle sum is $-180°$ (odd multiple with $l = 0$), so $s_0$ lies on the root locus and some $K$ meets the transient requirement.

**MARKER'S NOTES**

- Show each calculated angle on the figure.
- Many students lost the mark for never stating $L(s) = G(s)$ and where it comes from.

**3.2** *Gain at $s_0$, then check the steady-state requirement.* [10]

1. **Magnitude condition.**

$$K = \frac{1}{|L(s_0)|} = |s_0|\,|s_0 + 2| = \sqrt{5} \cdot \sqrt{5} = 5$$

2. **Ramp error by FVT.**

$$e_{ss} = \lim_{s \to 0} s \, \frac{1/s^2}{1 + \frac{K}{s(s+2)}} = \frac{2}{K} = \frac{2}{5} = 0.4$$

3. **Compare**

(any one of three equivalent checks): required $e_{ss} = 1/K_v = 0.01$ needs $K = 200$; or $0.4 \neq 0.01$; or $K_v = K/2 = 2.5 \neq 100$.

**ANSWER**

$K = 5$ gives $e_{ss} = 0.4$. Proportional control alone does not meet the steady-state requirement.

**3.3** *Lag compensator $D(s) = K\,\dfrac{s+z}{s+p}$.* [10]

1. **Know what $\alpha$ does.**

The lag network multiplies the low-frequency gain by $\alpha = z/p$ while leaving the locus near $s_0$ almost unchanged:

$$e_{ss} = \frac{2}{\alpha K} = 0.01, \quad K = 5 \;\Rightarrow\; \alpha = 40$$

2. **Place the zero well below $\omega$**

2 **Place the zero well below $\omega_n$**

(rule of thumb). At $s_0$, $\omega_n = |s_0| = \sqrt{5} \approx 2.24$ rad/s. A decade below gives

$$z \approx 0.224, \qquad p = \frac{z}{\alpha} \approx 0.0056$$

3 **Substitute.**

$$D(s) = 5\,\frac{s + 0.224}{s + 0.0056}$$

**ANSWER**

$\alpha = 40$, $z \approx 0.224$, $p \approx 0.0056$: $D(s) = 5\,(s + 0.224)/(s + 0.0056)$.

**MARKER'S NOTES**

- Two rule-of-thumb options were accepted; the point was to place the pair much lower than the natural frequency.

| ZERO, POLE | ANGLE ADDED AT $s_0$ | NEW DOMINANT POLES | VERDIC |
|---|---|---|---|
| $z = 0.224,\ p = 0.0056$ (one decade) | −5.2° | −0.88 ± j1.95 | Good |
| $z = 0.02,\ p = 0.0005$ (two decades) | −0.4° | −0.99 ± j2.00 | Good, slow ta |
| $z = 0.8,\ p = 0.02$ (other transcription) | −20.4° | −0.51 ± j1.94 | Too close to $s_0$ |

## RECONCILING THE TWO TRANSCRIPTIONS

Where the two versions disagreed, the value kept is the one that agrees with the audio and with the maths.

| ITEM | VERSION 1 | VERSION 2 | KEPT | REASON |
|---|---|---|---|---|
| Q1 specs | Inferred | Heard at the start | $e_{ss} = 0$, $T_s < 1$ s | Version 2 confirms the inference |
| Q2(a) marks | Not heard | 6 | 6 | Gives 26 + 26 + 28 = 80 |
| $\phi_{max}$ breakdown | 75° | 65° + 0° + 10° | Version 2 | 5% overshoot gives PM |

| | | | | |
|---|---|---|---|---|
| | | | | 64.6° |
| Lead $T$ | 0.51 s | 0.01 s | **0.51 s** | $1/(14\sqrt{0.02}) = 0.505$; 0.0102 is $\alpha T$ |
| Q3 $\omega_n$ | 2.24 rad/s | 3.24 rad/s | **2.24** | $K = 5$ gives $s^2 + 2s + 5$, $\omega_n = \sqrt{5}$ |
| Q3 plant and $s_0$ | $1/(s(s+2))$, $-1 \pm j2$ | Gain and $s_0$ left open | **Version 1** | The only fit for $K = 5$ and $e_{ss} = 2/K$ |
| Lag pole, zero | 0.224, 0.0056 | 0.8, 0.02 or 0.02, 0.0005 | **0.224, 0.0056** | 0.8 shifts the poles (table above) |
| Sub-part marks, time, header | Not stated | Splits such as 5/4/4/3/4 | **Dropped** | Not in the recording |

Reconstructed from the recorded memo walkthrough. Question 2's plant was only given as a Bode plot, so its readings (crossover about 14 rad/s, phase near −180°) are taken from the lecturer's spoken values.